

Database model documentation

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1. Model details

Model name:

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Version:

2.4

Database engine:

MySQL

Description:

Este modelo de base de datos está diseñado para un sistema de ventas y control de inventario, donde se gestionan clientes, socios, productos, ventas y el stock disponible. Su objetivo principal es permitir el registro, control y seguimiento de las operaciones comerciales de un negocio de forma organizada y eficiente.

2. Tables

2.1. Table PARTNER

Description:

This table is well structured and represents the people who work or participate in the business.

2.1.1. Columns

Column name	Type	Properties	Description
partner_id	int	PK	Unique partner identifier. Correct as primary key.
first_name	varchar(50)		Stores the partner's name. This is required.
apellido	varchar(50)		Save the partner's last name. Correct and required.
customer_address	varchar(15)		Identity document. It must be unique to avoid duplicates.
phone	varchar(20)	null	Contact number. This may be optional depending on the business.
role_id	int		Define the partner's role (e.g., administrator, salesperson). It is recommended to monitor their values.
ROL_id_rol	int		as INT PK is suitable for unique identification.

2.2. Table CLIENT

Description:

Suitable for storing customer information.

2.2.1. Columns

Column name	Type	Properties	Description
client_id	int	PK	Identificador único del cliente. Correcto como clave primaria
first_name	varchar(50)		Customer name.

			Required field.
last_name	varchar(50)		Customer's last name. Correct.
dni	varchar(15)		Customer document. It must be unique.
phone	varchar(20)		Customer contact. Important for communication.
address	varchar(100)		Customer location. Useful for shipping.
city	varchar(50)		Stores the name of the city where the client resides. It is used for location tracking and segmentation of clients.

2.3. Table PRODUCT

Description:

A well-designed table for products.

2.3.1. Columns

Column name	Type	Properties	Description
product_id	int	PK	Identificador único del producto.
product_name	varchar(100)		Nombre del producto. Obligatorio.
descripcion	varchar(255)	null	Detalle del producto. Opcional pero recomendable.
price	decimal(10,2)		Precio del producto. Correcto en formato decimal.
category_id	int		
unit_id	int		
CATEGORY_id_category	int		as INT PK is correct to uniquely identify each category.
UNIT_OF_MEASURE_unit_id	int		INT PK allows each unit to be identified uniquely.
INVENTORY_id_inventory	int		Inventory record identifier.
PURCHASE_id_purchase	int		

2.4. Table INVENTORY

Description:

Correct for stock control.

2.4.1. Columns

Column name	Type	Properties	Description
inventory_id	int	PK	Inventory record identifier.
product_id	int		Relationship to the product. Correct as a foreign key.
quantity_available	int		Current product stock.
updated_at	date		Date of last inventory change.
PURCHASE_id_purchase	int		

2.5. Table SALE

Description:

Very well planned, it is the core of the system.

2.5.1. Columns

Column name	Type	Properties	Description
sale_id	int	PK	Unique sale identifier.
sale_date	date		Date the sale takes place
total_amount	decimal(10,2)		Total sale amount.
partner_id	int		Selling Partner (FK).
client_id	int		Buying Customer (FK).
PRODUCT_id_product	int		Unique product identifier.
PARTNER_id_partner	int		Unique partner identifier. Correct as primary key.
CLIENT_client_id	int		Unique customer identifier. Correct as a primary key

2.6. Table SALE_DETAIL

Description:

Correcta para descomponer cada venta en productos.

2.6.1. Columns

Column name	Type	Properties	Description
detail_id	int	PK	Detail identifier (optional if using a composite key).
sale_id	int		Relationship with sales.
product_id	int		Product sold.
quantity	int		Quantity sold.
unit_price	decimal(10,2)		Price of the product at that time.
SALE_id_sale	int		Unique sale identifier.
PRODUCT_id_product	int		Unique product identifier.

2.7. Table ROLE

Description:

The ROL entity is well designed and standardized, as it correctly separates the concept of partner role, avoiding storing repeated text in other tables.

2.7.1. Columns

Column name	Type	Properties	Description
role_id	int	PK	como INT PK es adecuado para identificación única.
role_name	varchar(30)	null	representa correctamente la descripción del rol.

2.8. Table UNIT_MEASURE

Description:

The UNIT_OF_MEASURE entity is correctly designed and represents a good standardization practice by separating the product's units of measure.

2.8.1. Columns

Column name	Type	Properties	Description
unit_id	int	PK	INT PK allows each unit to be identified uniquely.
unit_name	varchar(20)		Describe the unit of measurement correctly.

2.9. Table CATEGORY

Description:

The CATEGORY entity is correctly structured and complies with a basic design suitable for a relational logical model.

2.9.1. Columns

Column name	Type	Properties	Description
category_id	int	PK	as INT PK is correct to uniquely identify each category.
category_name	varchar(50)		It correctly represents the category description.

2.10. Table PURCHASE

Description:

The PURCHASE table is used to record product purchases made by the Tuta Wayta system from suppliers.

2.10.1. Columns

Column name	Type	Properties	Description
id_purchase	int	PK	Unique identifier for each purchase registered in the system.
id_product	int		Relates the purchase to the purchased product through a foreign key.
purchase_date	date		Stores the date when the product purchase was made.
quantity	int	null	Records the quantity of purchased products and validates that it is greater than zero.
purchase_price	decimal(10,2)		Stores the purchase price of the product and validates that it is greater than zero.
id_partner	int		Relates the purchase to the responsible partner through a foreign key.

supplier	varchar(100)		Stores the name of the supplier providing the product.
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2.11. Table USER

Description:

It serves to store the system access information of the partners registered in the project

2.11.1. Columns

Column name	Type	Properties	Description
id_user	int	PK	
id_partner	int	PK	
username	varchar(50)		
password	varchar(255)		
last_login	timestamp		
PARTNER_2_partne r_id	int		

3. References

3.1. Reference SALE_DETAILS_SALE

Description:

This table typically represents a relationship between a sale and its details, but the name suggests it's duplicated or misnamed. In correct models, the standard practice is to use SALE_DETAIL.

SALE	0..*	SALE_DETAIL
sale_id	<->	SALE_id_sale

3.2. Reference PRODUCT_SALE

Description:

Entity that links sales to products sold, allowing the quantity and price of each product within a sale to be stored.

PRODUCT	0..*	SALE
product_id	<->	PRODUCT_id_product

3.3. Reference PRODUCT_CATEGORY

Description:

Entity that links products to their respective categories, allowing products to be classified within the system.

CATEGORY	0..*	PRODUCT
category_id	<->	CATEGORY_id_category

3.4. Reference PRODUCT_UNIT_MEASURE

Description:

Entity that relates products to their unit of measurement, defining how each product is quantified (for example: unit, kilogram, liter).

UNIT_MEASURE	0..*	PRODUCT
unit_id	<->	UNIT_OF_MEASURE_unit_id

3.5. Reference INVENTORY_PRODUCT

Description:

Entity that stores product inventory information, such as quantity available, entries, exits and

stock control.

INVENTORY	0..*	PRODUCT
inventory_id	<->	INVENTORY_id_inventory

3.6. Reference PARTNER_SALE

Description:

Intermediate table that manages the relationship between PARTNER and SALE, recording the commercial operations carried out by each partner for the purposes of control, monitoring and sales analysis.

PARTNER	1..*	SALE
partner_id	<->	PARTNER_id_partner

3.7. Reference CUSTOMER_SALE

Description:

Entity that links the sale to the customer who makes it, allowing identification of which customer each transaction belongs to.

CLIENT	0..*	SALE
client_id	<->	CLIENT_client_id

3.8. Reference PARTNER_ROLE

Description:

Table that records the assignment of roles to partners within the system, allowing the definition of the permissions or responsibilities that each partner has.

ROLE	0..*	PARTNER
role_id	<->	ROL_id_rol

3.9. Reference PRODUCT_SALE_DETAILS

Description:

Entity that stores the details of the products associated with a sale, including quantity, price and subtotal for each item sold.

PRODUCT	0..*	SALE_DETAIL
product_id	<->	PRODUCT_id_product

3.10. Reference INVENTORY_PURCHASE

Description:

The INVENTARIO_COMPRA table is used to manage and record the relationship between inventory and product purchases in the Tuta Wayta system.

PURCHASE	0..*	INVENTORY
id_purchase	<->	PURCHASE_id_purchase

3.11. Reference PRODUCT_PURCHASE

Description:

The PRODUCTO_COMPRA table is used to store the relationship between purchased products and purchase records in the Tuta Wayta system

PURCHASE	1..*	PRODUCT
id_purchase	<->	PURCHASE_id_purchase

3.12. Reference USER_PARTNER

Description:

The USER_MEMBER table is used to link user information with members registered in the system.

PARTNER	0..*	USER
partner_id	<->	PARTNER_2_partner_id

4. Areas

4.1. Area_1 subject area

4.1.1. Tables

- PARTNER
- ROLE
- USER

4.1.2. References

- SALE_DETAILS_SALE
- PRODUCT_SALE
- PRODUCT_CATEGORY
- PRODUCT_UNIT_MEASURE
- INVENTORY_PRODUCT
- PARTNER_SALE
- CUSTOMER_SALE
- PARTNER_ROLE
- PRODUCT_SALE_DETAILS
- INVENTORY_PURCHASE
- PRODUCT_PURCHASE
- USER_PARTNER

4.2. Area_2 subject area

4.2.1. Tables

- CLIENT
- PRODUCT
- SALE
- SALE_DETAIL

4.2.2. References

- SALE_DETAILS_SALE
- PRODUCT_SALE
- PRODUCT_CATEGORY
- PRODUCT_UNIT_MEASURE
- INVENTORY_PRODUCT
- PARTNER_SALE
- CUSTOMER_SALE
- PARTNER_ROLE
- PRODUCT_SALE_DETAILS
- INVENTORY_PURCHASE
- PRODUCT_PURCHASE
- USER_PARTNER

4.3. Area_3 subject area

4.3.1. Tables

- INVENTORY
- UNIT_MEASURE
- CATEGORY
- PURCHASE

4.3.2. References

- SALE_DETAILS_SALE
- PRODUCT_SALE
- PRODUCT_CATEGORY
- PRODUCT_UNIT_MEASURE
- INVENTORY_PRODUCT
- PARTNER_SALE
- CUSTOMER_SALE
- PARTNER_ROLE
- PRODUCT_SALE_DETAILS
- INVENTORY_PURCHASE
- PRODUCT_PURCHASE
- USER_PARTNER